

BALA MUHAMMAD MUHAMMAD

D52 Sabon Layi Gangare, Jos, Plateau State, Nigeria

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EDUCATION AND TRAINING

Master of Science (M.Sc.) in Mathematics

Aug 2024 – May 2026

Universitas Andalas, Padang, Indonesia | CGPA: 3.63 / 4.00 | EQF Level 7

- Thesis: Stability and Bifurcation Analysis of a Bird Flu Model with Vaccination in a Poultry Farm

Bachelor of Science (Honours) in Computer Science

Mar 2018 – Jan 2024

Aliko Dangote University of Science and Technology (ADUST), Wudil, Kano, Nigeria | CGPA: 4.07 / 5.00 (Upper Second Class)

| EQF Level 6

- Thesis: Design and Implementation of a Web-Based Information System for University Staff Union

RESEARCH EXPERIENCE AND PROJECTS

Graduate Researcher – MSc Thesis

Sep 2025 – Mar 2026

Department of Mathematics, Universitas Andalas, Padang, Indonesia

- Modelled avian influenza transmission dynamics, deriving the basic reproduction number (R_0) and uncovering critical stability and bifurcation behaviour to optimise vaccination as a disease-control strategy.

Undergraduate Research Assistant

Feb 2023 – Aug 2023

Department of Computer Science, ADUST, Wudil, Nigeria

- Designed and developed the ASUU Web-Based Information System (AWBIS), a full-stack platform (HTML, CSS, JavaScript, PHP, MySQL) digitising member management, communication and biometric attendance for a university staff union; built through stakeholder requirements-gathering and Waterfall SDLC, and formally adopted by ASUU ADUST Wudil.

Independent Research Projects

- Automated Numerical Method Selection via Deterministic Finite Automata: DFA-based compiler framework (Python) that classifies expressions and auto-selects numerical solvers with high accuracy and low overhead.
- Automated Attendance System Using Face Recognition: Viola-Jones, Retinex and PCA; over 90% accuracy.
- Secure ID Request and Administration System, ADUST Wudil: web-based system digitising university ID card requests (Agile, PHP/MySQL).
- Web-Based Personalised Dietary Support System for Diabetic Patients: PHP/MySQL meal-recommendation system validated with 86 diabetic patients; high satisfaction and usability ratings.
- Hospital Emergency Response Management System (HERMS): web app for real-time accident reporting and hospital-side emergency response coordination using geospatial mapping.

SKILLS

- Programming:** Java, Python, C, C#, JavaScript, SQL, TypeScript, HTML/CSS, LaTeX
- ML & Data Science:** TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas, OpenCV
- Web Development:** Spring Boot, Angular, Node.js, React, Flask, Django
- Databases:** MySQL, MongoDB, Oracle, PostgreSQL, DynamoDB, Redis
- DevOps & Cloud:** Docker, Kubernetes, AWS, Azure, GCP, Git, Terraform
- Networking & Simulation:** Cisco Packet Tracer, Wireshark
- GIS & Geospatial:** ArcGIS, QGIS
- Testing:** JUnit, Mockito, Selenium, Cypress
- Analytics & Visualisation:** Tableau, Matplotlib, Seaborn, Power BI, SAS Studio
- Citation Management:** Mendeley, Zotero
- Soft Skills:** Problem solving, leadership, team collaboration, communication, time management

PUBLICATIONS

- Bala M.M. (2026 – Under Review).** Trust-Aware Federated Learning for Heart Disease Prediction with Integrated Verification, Dynamic Trust Modeling, and Blockchain-Based Auditability. SN Computer Science (Springer).
- Muhammad, B.M. & Bahri, S. (2026).** Automating Numerical Method Selection Using Deterministic Finite Automata in a Compiler-Based Framework. Journal of Computational Innovation and Analytics, 5(2).
- Muhammad, B.M., Lawan, A.A. & Abdi, A.S. (2026).** Harnessing Machine Learning to Forecast Social Media Advocacy Success (ICOASE2025). Al-Rafidain Journal of Computer Sciences and Mathematics, 20(1), 53–63.

- **Abubakar, S., Dauda, M.K., Musa, M.A. & Muhammad, B.M. (2026).** Design and Implementation of a Web-Based Lecture Timetable Scheduling System. *Journal of Informatics and Web Engineering*, 5(1).
- **Muhammad, B.M., Lawan, A.A., Bala, J., Abdulrauf, T.S. & Bala, I.M. (2025).** Predictive Modeling of Student Career Pathways Using Machine Learning Techniques. *Journal of Statistical Sciences and Computational Intelligence*, 1(3), 166–174.
- **Muhammad, B.M., Lawan, A.A. & Abdulrahman, S.M. (2024).** Design and Implementation of a Web-Based Information System for University Staff Union. *Journal of Computational Innovation and Analytics*, 3(2), 89–109.
- **Bala, I.M. et al. (2024).** Smart Technology Integration: SHM and BIM for Preventing Building Collapses. *African Journal of Advances in Science and Technology Research*, 17(1).
- **Tidiane, G. & Muhammad, B.M. (2024).** Investigating the Long-run Relationship between Public Debt and Economic Growth: Evidence from Mali. *Frontiers in Business and Economics*, 3(3), 199–210.

CONFERENCES, SEMINARS, AND WORKSHOPS

Presenter, 5th International Conference on Advanced Science and Engineering (ICOASE 2025) 24 – 25 Sep 2025

- Presented "Harnessing Machine Learning to Forecast Social Media Advocacy Success", applying five ML models to predict advocacy campaign success with up to 98% accuracy (Random Forest).

4th International School on Modeling and Analytics (Summer Course), Federal University Oye-Ekiti Sep 2026

- Attended a three-week international summer course on modeling and analytics, delivered virtually.

International Summer Course on Ecotourism, Nature Conservation & Food Security, IPB University, Indonesia Jul – Nov 2025

- Completed a 4-credit international programme covering biodiversity, sustainable ecosystems and food security.

NUNI International Seminar Series #10 (Virtual): Educational Initiatives Supporting the SDGs in Vietnam 25 Nov 2025

Professional Seminars in Research and Statistical Methods, Instats

- StatKey; Principles of Research Design (0.5 ECTS) (Oct 2025).
- Science Writing and Storytelling (Oct 2025).
- ASSON 2025 National Statistics Student Conference (Oct 2025).

Citation and References, Elsevier Researcher Academy (in person), ADUST Wudil 30 Nov 2022

Professional Workshops

- Research Methodology and Scientific Writing, MAAUN (Oct 2025).
- Identifying Research Gaps and Researchable Areas in Computing, Kano State Young Researchers Forum (Feb 2026).
- UNESCO International Day of Light, Photonics Nigeria (May 2026).
- Defining Research and Innovation Strategy for Excellence (Oct 2025).

WORK EXPERIENCE

Customer Support Intern (Remote) Sep 2026 – Present

Zivo, Yaba, Lagos, Nigeria

- Provide frontline customer support via live chat and email, resolving product, service and order-related inquiries and complaints in line with service-level expectations.
- Manage customer tickets, document interactions and escalate unresolved technical issues to internal teams to improve response quality and customer satisfaction.

Field Data Officer & Technical Enumerator Jun 2026 – Sep 2026

ADATECH Installers, Kano State, Nigeria

- Executed structured field data collection (GPS coordinates, geotagged imagery, customer and technical records) via mobile data capture tools, in full compliance with MOJEC/KEDCO data standards.
- Validated meter installation data for accurate field-to-database synchronisation, delivered on-site technical training to customers, and upheld data quality control across assigned enumeration zones.

IT Support Technician & Computer Trainer (Contract) Feb 2024 – Jul 2024

De-Khalid Computer Institute, Jos, Nigeria

- Provided technical support across computer labs, troubleshooting hardware, software and network issues, and performed system maintenance including OS installation, software updates and lab readiness checks.
- Configured and maintained LAN connections and peripherals (printers, projectors), documenting recurring faults to improve maintenance turnaround and reliability.
- Delivered computer literacy and Computer-Based Test (CBT) readiness training to students, job applicants and staff for platforms used in JAMB and job-screening assessments.

Industrial Trainee – Software Development & Networking

Sep 2021 – Mar 2022

Black Innovations Africa (BIA), Jos, Nigeria

- Completed a 6-month industrial training (SIWES) as part of the BSc requirement, with exposure to software development and networking operations.
- Assisted senior developers across the SDLC, contributing to coding, debugging and testing of web applications using PHP, JavaScript, MySQL and Git.
- Supported networking operations including router and switch configuration, structured cabling, IP addressing and subnetting, and troubleshooting with Wireshark and Packet Tracer.

TEACHING & MENTORING EXPERIENCE

Mathematics & Computer Studies Teacher | Examination Officer (Full-Time)

Sep 2026 – Present

Al-Hilal Secondary School, Jos, Plateau State, Nigeria

- Teach Computer Studies and Mathematics, and serve as Examination Officer, managing exam planning, timetabling, invigilation and result processing.

Computer Technology Instructor

May 2022 – Nov 2022

Skynet Information Technology Institute, Jos, Nigeria

- Designed and delivered a structured curriculum in Data Management and Data Analysis, training students in MySQL, Excel, Tableau and Power BI through theory and hands-on labs.
- Taught network simulation and configuration with Cisco Packet Tracer (LAN/WAN architecture, IP addressing, routing, switching) alongside foundational IT literacy.
- Developed lesson plans and assessments, supervised practical labs and provided one-on-one academic support.

Peer Tutor & Academic Mentor, Faculty of Computing and Mathematical Sciences, ADUST Wudil Jul 2019 – Jan 2024

- Tutored 100- and 200-level students in Mathematics and Statistics, and coursemates in Algorithm Development & Analysis, Data Structures and Net-Centric Computing.
- Helped coursemates summarise and interpret complex lecture material, and mentored junior students on study strategies and academic guidance.

MASSIVE OPEN ONLINE COURSES (MOOCS)

- GIS Basic Course (GIS: Fundamentals to Advanced Applications), Precision Field Academy (Sep 2026)
- IoT Wireless & Cloud Computing Emerging Technologies, Yonsei University (Sep 2026)
- Designing and Formatting a Presentation in PowerPoint, Coursera / MTN (Sep 2026)
- Quantum Computing & Quantum Machine Learning: Advanced, Real Numbers (Sep 2026)
- Introduction to Greenhouse Gas Accounting for IT, Cisco Networking Academy (Sep 2026)
- Introduction to Cybersecurity; Network Support and Security, Cisco Networking Academy (Aug 2026)
- Integrated Photonic Circuit Design, Udemy (Aug 2026)
- Introduction to IoT and Digital Transformation, Cisco Networking Academy (Jul 2026)
- Biodiversity Finance: Designing and Implementing Finance Plans for Nature, BIOFIN / UNDP (Oct 2025)
- Introduction to Python OpenCV, Simplilearn | SkillUp (Oct 2025)
- Sell Domains And Web Hosting Under Your Name, Udemy (Oct 2025)
- The Data Analyst Course: Complete Data Analyst Bootcamp, Udemy / 365 Careers (Oct 2025)
- Data Science Marathon: 120 Projects To Build Your Portfolio, Udemy / Pianalytix (Oct 2025)
- Data Science: Deep Learning and Neural Networks in Python, Udemy (May 2025)
- Deep Learning Bootcamp: PyTorch, TensorFlow & Deployment, Udemy (Jun 2024)
- Microsoft Excel 2019 / 365, Earn and Excel (Oct 2023)
- Clean-IT: Towards Sustainable Digital Technologies, OPEN HPI (Jul 2023)
- Sustainable Software Engineering, OPEN HPI (Feb 2023)

LEADERSHIP EXPERIENCE

President, Jos Students' Consultative Forum (JOSCOF), ADUST Wudil

2022 – 2023

- Elected by the student body to represent student interests at union congress and executive meetings.

Treasurer, Independent Electoral Committee, Association of Computing and Mathematical Science Students, ADUST Wudil

2022 – 2023

- Nominated to manage financial budgeting and reporting for the successful 2023 student elections.

Welfare Officer, National Association of Computer Science Students (NACOSS), ADUST Wudil Chapter 2022 – 2023

- Nominated to address student welfare concerns and liaise between the student body and departmental leadership.

VOLUNTEER EXPERIENCE

State Coordinator, Plateau State, Youth Coalition Against Cancer (YOCAC)

Sep 2026 – Present

- Lead cancer awareness, prevention, advocacy and support initiatives across Plateau State, contributing to the organisation's national mission on cancer education and research.

SDGs Community Development Service (CDS), National Youth Service Corps (NYSC), Jos, Nigeria Feb 2024 – Feb 2025

- Participated in SDG-aligned community development initiatives, supporting awareness campaigns and grassroots projects in education, health and environmental sustainability.

Public Health Volunteer, Primary Healthcare Center, Gangare Ward (COVID-19 Response)

2020

- Supported community health outreach, including door-to-door distribution of anti-malarial drugs and mosquito nets to underserved households.
- Assisted nurses in child immunisation drives and public health awareness campaigns on disease prevention, hygiene and COVID-19 safety.

HONOURS AND AWARDS

Postgraduate Scholarship Award, Universitas Andalas, Indonesia

17 Jul 2024

- Full scholarship covering tuition, accommodation, living expenses, health insurance, language course, visa and travel.

Dean's List Award for Academic Excellence, Department of Computer Science, ADUST Wudil

16 Jan 2024

- Graduated in the top 3% of the class with a CGPA of 4.07/5.00.

Award of Excellence in Leadership, JOSCOF ADUST Wudil Chapter

15 Dec 2023

ASUU Award for Excellence – AWBIS Project, ASUU ADUST Wudil Chapter

2023

- Cash award for designing and developing the ASUU Web-Based Information System, later adopted for real-world use.

Kabiru Custom Foundation Educational Support Award

6 Mar 2019

- Monthly educational stipend for academic performance and merit, sustained over four years.

NETWORKS AND MEMBERSHIPS

- Royal Statistical Society (RSS) – Member
- American Statistical Association (ASA) – Student Member
- American Society for Microbiology (ASM) – Global Outreach Contributing Member, Computational Microbiology and Modeling
- American Society of Civil Engineers (ASCE) – Student Member
- AlHuda Centre of Islamic Banking and Economics (CIBE), UAE – Global Fellow
- Institute for Engineering Research and Publications (IFERP) – Student Member

LANGUAGE SKILLS

Mother tongue: Hausa (native)

Other languages: English (fluent); Bahasa Indonesia (basic); Arabic (basic)